

EECE7352 Computer Architecture

Homework 1

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Part C:

I have chosen the **STREAM** benchmark for my analysis.

STREAM BENCHMARK:

STREAM is a synthetic benchmark that measures **sustainable memory bandwidth** (MB/s) and the corresponding computation rate for simple vector kernels.

- It uses four core operations on long vectors:

- **Copy:** $a[i] = b[i]$
- **Scale:** $a[i] = q * b[i]$
- **Add:** $a[i] = b[i] + c[i]$
- **Triad:** $a[i] = b[i] + q * c[i]$

Each kernel stresses the **memory subsystem** (main memory, caches, memory controller, and interconnect), making STREAM an excellent test of how efficiently a system can move data, not just compute.

Run Results

X86_64 (O0 - O3)

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lota.coe.neu.edu_gcc_O0.txt X
PartC > runs > lota.coe.neu.edu_gcc_O0.txt
1 -----
2 STREAM version $Revision: 5.10 $
3 -----
4 This system uses 8 bytes per array element.
5 -----
6 Array size = 10000000 (elements), Offset = 0 (elements)
7 Memory per array = 76.3 MiB (= 0.1 GiB).
8 Total memory required = 228.9 MiB (= 0.2 GiB).
9 Each kernel will be executed 10 times.
10 The *best* time for each kernel (excluding the first iteration)
11 will be used to compute the reported bandwidth.
12 -----
13 Your clock granularity/precision appears to be 1 microseconds.
14 Each test below will take on the order of 32840 microseconds.
15 (= 32840 clock ticks)
16 Increase the size of the arrays if this shows that
17 you are not getting at least 20 clock ticks per test.
18 -----
19 WARNING -- The above is only a rough guideline.
20 For best results, please be sure you know the
21 precision of your system timer.
22 -----
23 Function Best Rate MB/s Avg time Min time Max time
24 Copy: 5361.2 0.029866 0.029844 0.029895
25 Scale: 5147.7 0.031107 0.031082 0.031135
26 Add: 6763.8 0.035504 0.035483 0.035531
27 Triad: 6053.9 0.039664 0.039644 0.039693
28 -----
29 Solution Validates: avg error less than 1.000000e-13 on all three arrays
30 -----
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lota.coe.neu.edu_gcc_O1.txt X
PartC > runs > lota.coe.neu.edu_gcc_O1.txt
1 -----
2 STREAM version $Revision: 5.10 $
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4 This system uses 8 bytes per array element.
5 -----
6 Array size = 10000000 (elements), Offset = 0 (elements)
7 Memory per array = 76.3 MiB (= 0.1 GiB).
8 Total memory required = 228.9 MiB (= 0.2 GiB).
9 Each kernel will be executed 10 times.
10 The *best* time for each kernel (excluding the first iteration)
11 will be used to compute the reported bandwidth.
12 -----
13 Your clock granularity/precision appears to be 1 microseconds.
14 Each test below will take on the order of 12125 microseconds.
15 (= 12125 clock ticks)
16 Increase the size of the arrays if this shows that
17 you are not getting at least 20 clock ticks per test.
18 -----
19 WARNING -- The above is only a rough guideline.
20 For best results, please be sure you know the
21 precision of your system timer.
22 -----
23 Function Best Rate MB/s Avg time Min time Max time
24 Copy: 10317.3 0.015529 0.015508 0.015553
25 Scale: 10360.0 0.015453 0.015444 0.015459
26 Add: 11192.5 0.021457 0.021443 0.021478
27 Triad: 11145.2 0.021544 0.021534 0.021558
28 -----
29 Solution Validates: avg error less than 1.000000e-13 on all three arrays
30 -----
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```

lota.coe.neu.edu_gcc_O2.txt X

PartC > runs > lota.coe.neu.edu_gcc_O2.txt

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STREAM version \$Revision: 5.10 \$

This system uses 8 bytes per array element.

Array size = 10000000 (elements), Offset = 0 (elements)
Memory per array = 76.3 MiB (= 0.1 GiB).
Total memory required = 228.9 MiB (= 0.2 GiB).
Each kernel will be executed 10 times.
The *best* time for each kernel (excluding the first iteration)
will be used to compute the reported bandwidth.

Your clock granularity/precision appears to be 1 microseconds.
Each test below will take on the order of 9999 microseconds.
(= 9999 clock ticks)
Increase the size of the arrays if this shows that
you are not getting at least 20 clock ticks per test.

WARNING -- The above is only a rough guideline.
For best results, please be sure you know the
precision of your system timer.

Function Best Rate MB/s Avg time Min time Max time
Copy: 11050.5 0.014494 0.014479 0.014506
Scale: 10485.6 0.015284 0.015259 0.015329
Add: 11337.4 0.021179 0.021169 0.021197
Triad: 11230.1 0.021405 0.021371 0.021485

Solution Validates: avg error less than 1.000000e-13 on all three arrays

lota.coe.neu.edu_gcc_O3.txt X

PartC > runs > lota.coe.neu.edu_gcc_O3.txt

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STREAM version \$Revision: 5.10 \$

This system uses 8 bytes per array element.

Array size = 10000000 (elements), Offset = 0 (elements)
Memory per array = 76.3 MiB (= 0.1 GiB).
Total memory required = 228.9 MiB (= 0.2 GiB).
Each kernel will be executed 10 times.
The *best* time for each kernel (excluding the first iteration)
will be used to compute the reported bandwidth.

Your clock granularity/precision appears to be 1 microseconds.
Each test below will take on the order of 9957 microseconds.
(= 9957 clock ticks)
Increase the size of the arrays if this shows that
you are not getting at least 20 clock ticks per test.

WARNING -- The above is only a rough guideline.
For best results, please be sure you know the
precision of your system timer.

Function Best Rate MB/s Avg time Min time Max time
Copy: 11052.0 0.014497 0.014477 0.014523
Scale: 10538.1 0.015197 0.015183 0.015216
Add: 11230.2 0.021395 0.021371 0.021448
Triad: 11185.2 0.021491 0.021457 0.021616

Solution Validates: avg error less than 1.000000e-13 on all three arrays

Arm_v8 (O0-O3)

degrees_gcc_O0.txt X

PartC > runs > degrees_gcc_O0.txt

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STREAM version \$Revision: 5.10 \$

This system uses 8 bytes per array element.

Array size = 10000000 (elements), Offset = 0 (elements)
Memory per array = 76.3 MiB (= 0.1 GiB).
Total memory required = 228.9 MiB (= 0.2 GiB).
Each kernel will be executed 10 times.
The *best* time for each kernel (excluding the first iteration)
will be used to compute the reported bandwidth.

Your clock granularity/precision appears to be 1 microseconds.
Each test below will take on the order of 76997 microseconds.
(= 76997 clock ticks)
Increase the size of the arrays if this shows that
you are not getting at least 20 clock ticks per test.

WARNING -- The above is only a rough guideline.
For best results, please be sure you know the
precision of your system timer.

Function Best Rate MB/s Avg time Min time Max time
Copy: 2074.4 0.077256 0.077130 0.077315
Scale: 2667.7 0.060042 0.059976 0.060097
Add: 3311.3 0.072537 0.072479 0.072624
Triad: 3218.1 0.074846 0.074578 0.075807

Solution Validates: avg error less than 1.000000e-13 on all three arrays

degrees_gcc_O1.txt X

PartC > runs > degrees_gcc_O1.txt

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STREAM version \$Revision: 5.10 \$

This system uses 8 bytes per array element.

Array size = 10000000 (elements), Offset = 0 (elements)
Memory per array = 76.3 MiB (= 0.1 GiB).
Total memory required = 228.9 MiB (= 0.2 GiB).
Each kernel will be executed 10 times.
The *best* time for each kernel (excluding the first iteration)
will be used to compute the reported bandwidth.

Your clock granularity/precision appears to be 1 microseconds.
Each test below will take on the order of 22577 microseconds.
(= 22577 clock ticks)
Increase the size of the arrays if this shows that
you are not getting at least 20 clock ticks per test.

WARNING -- The above is only a rough guideline.
For best results, please be sure you know the
precision of your system timer.

Function Best Rate MB/s Avg time Min time Max time
Copy: 7253.3 0.022153 0.022059 0.022271
Scale: 6825.3 0.023515 0.023442 0.023620
Add: 6620.5 0.036355 0.036251 0.036455
Triad: 6559.9 0.036691 0.036586 0.036840

Solution Validates: avg error less than 1.000000e-13 on all three arrays

degrees_gcc_O2.txt X

PartC > runs > degrees_gcc_O2.txt

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STREAM version \$Revision: 5.10 \$

This system uses 8 bytes per array element.

Array size = 10000000 (elements), Offset = 0 (elements)
Memory per array = 76.3 MiB (= 0.1 GiB).
Total memory required = 228.9 MiB (= 0.2 GiB).
Each kernel will be executed 10 times.
The *best* time for each kernel (excluding the first iteration)
will be used to compute the reported bandwidth.

Your clock granularity/precision appears to be 1 microseconds.
Each test below will take on the order of 22485 microseconds.
(= 22485 clock ticks)
Increase the size of the arrays if this shows that
you are not getting at least 20 clock ticks per test.

WARNING -- The above is only a rough guideline.
For best results, please be sure you know the
precision of your system timer.

Function Best Rate MB/s Avg time Min time Max time
Copy: 7251.6 0.022109 0.022064 0.022154
Scale: 6908.5 0.023184 0.023160 0.023206
Add: 6623.2 0.036293 0.036236 0.036387
Triad: 6596.5 0.036478 0.036383 0.036555

Solution Validates: avg error less than 1.000000e-13 on all three arrays

degrees_gcc_O3.txt X

PartC > runs > degrees_gcc_O3.txt

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STREAM version \$Revision: 5.10 \$

This system uses 8 bytes per array element.

Array size = 10000000 (elements), Offset = 0 (elements)
Memory per array = 76.3 MiB (= 0.1 GiB).
Total memory required = 228.9 MiB (= 0.2 GiB).
Each kernel will be executed 10 times.
The *best* time for each kernel (excluding the first iteration)
will be used to compute the reported bandwidth.

Your clock granularity/precision appears to be 1 microseconds.
Each test below will take on the order of 21089 microseconds.
(= 21089 clock ticks)
Increase the size of the arrays if this shows that
you are not getting at least 20 clock ticks per test.

WARNING -- The above is only a rough guideline.
For best results, please be sure you know the
precision of your system timer.

Function Best Rate MB/s Avg time Min time Max time
Copy: 7629.6 0.021040 0.020971 0.021197
Scale: 7671.0 0.020912 0.020858 0.020951
Add: 6799.4 0.035362 0.035297 0.035425
Triad: 6766.5 0.035506 0.035469 0.035628

Solution Validates: avg error less than 1.000000e-13 on all three arrays

Observation

On x86-64 (Iota/Zeta)

- At -O0, bandwidth is very low: Copy $\approx$ **5961 MB/s**, Triad $\approx$ **6053 MB/s**.
- At -O1, performance jumps dramatically: Copy $\approx$ **10317 MB/s**, Triad $\approx$ **11192 MB/s**.
- At -O2, results improve further: Copy $\approx$ **11056 MB/s**, Triad $\approx$ **11230 MB/s**.
- At -O3, the peak is reached: Copy $\approx$ **11502 MB/s**, Triad $\approx$ **11185 MB/s**.

Optimizations enabled vectorization and better memory access patterns, roughly **doubling performance** from -O0 to -O3.

On ARMv8 (degrees)

- At -O0, bandwidth is much lower: Copy $\approx$ **2074 MB/s**, Triad $\approx$ **3218 MB/s**.
- At -O1, performance improves strongly: Copy $\approx$ **7253 MB/s**, Triad $\approx$ **6599 MB/s**.
- Higher optimization levels (-O2, -O3) continue improving but are still lower than x86-64.

ARMv8 shows a **3–4 times the gain from compiler optimizations**, but its absolute memory bandwidth is lower than the x86-64 node in this cluster.

Analysis of Results

1. Impact of Compiler Optimizations

- The leap from -O0 $\rightarrow$ -O1 shows how critical loop optimizations and vectorization are for STREAM.
- At -O0, memory accesses are scalar and unoptimized, leading to very poor utilization of memory bandwidth.
- At -O2/-O3, vectorization and prefetching allow the compiler to issue wider memory operations and overlap loads/stores, significantly increasing throughput.

2. x86-64 vs ARMv8

- x86-64 delivers peak **$\sim$ 11.5 GB/s**, ARMv8 peaks around **$\sim$ 7.2 GB/s**
- This gap reflects differences in memory subsystem design:
 - x86 server CPUs often have **wider memory buses** and **higher clocked DRAM channels**.
 - ARM cores here are likely tuned for energy efficiency, not raw memory bandwidth.

Suggested System for Better Efficiency

The **STREAM benchmark** would run more efficiently on a high-bandwidth memory (HBM)–equipped system, e.g.:

- **Modern GPUs** such as NVIDIA A100, AMD MI250 etc
- HPCs like Fugaku supercomputers (Fujitsu A64FX) provide >900 GB/s of sustained memory bandwidth.